TYNDP // 2024

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Annex to the Final Scenarios Report

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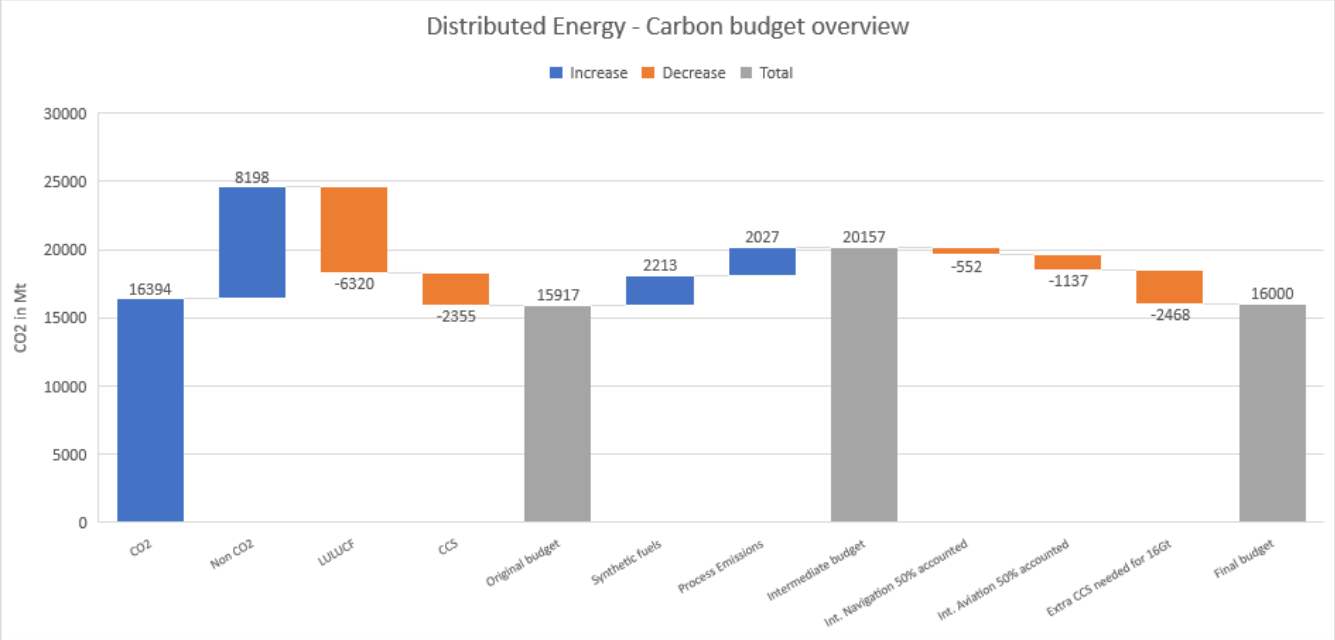
After receiving the feedback from the European Scientific Board on Climate Change (ESABCC) on the Draft TYNDP 2024 scenario report, ENTSO-E and ENTSOG have revised the carbon budget calculation.

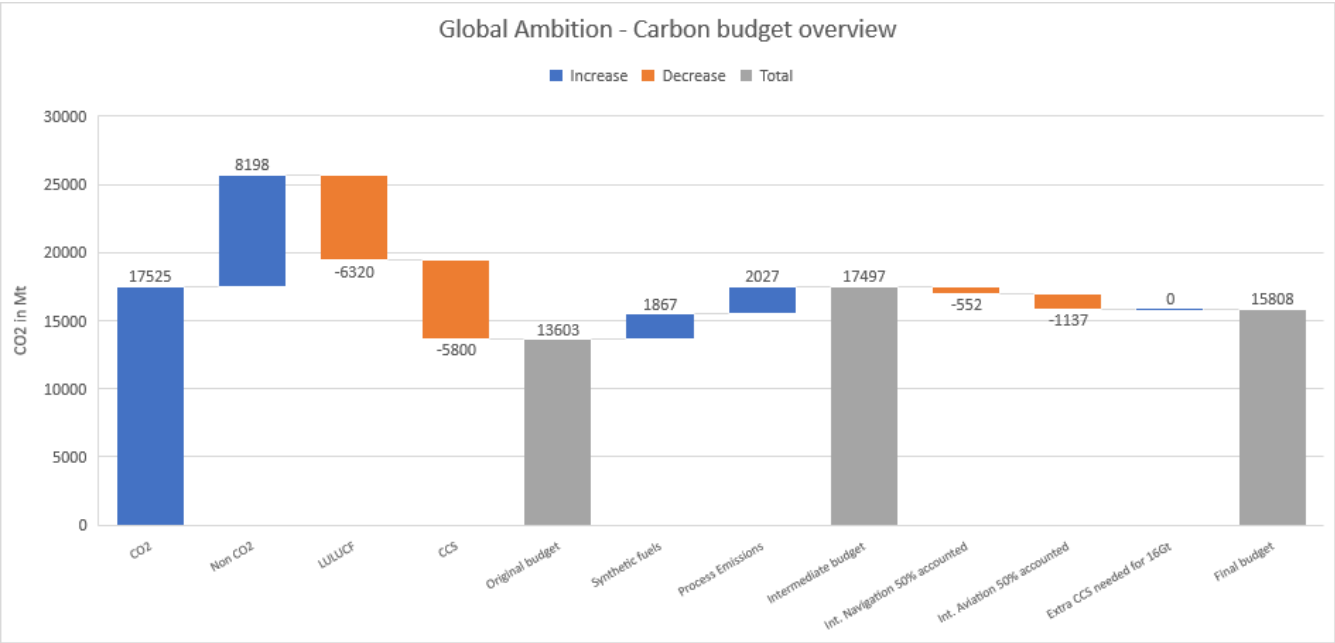
The revision includes an update on how the CCU/S related emissions, process emissions and CO2 emissions related to the international transport sector are taken into account for the carbon budget calculation.

The approach taken is outlined below:

- CO2 emissions were the starting point. The non-CO2 emissions were added, LULUCF and CCS are taken into account as negative emissions. The result is the "Original Budget" as published.
- The CO2 emissions from synthetic fuels were added to amend the calculation regarding CCS/U.
- The CO2 process emissions (e.g. cement and lime) as estimated by ESABCC of 2Gt CO2 were added. This leads to an "Intermediate budget".
- The assumptions on international aviation and international navigation CO2 emission accounting were revised.
    - The emissions of these two sectors were fully accounted for in the original calculation. This was considered a conservative approach.
    - The updated approach follows the more common approach, taking 50% of international navigation and 50 % of international aviation into account.
- An assessment of the carbon budget to stay with the 16Gt carbon budget threshold for upper limit shows that
    - 2.468 MT CO2 extra CCS is required cumulative from 2030 to 2050 for the Distributed Energy Scenario to stay within the 16Gt Co2 carbon budget
    - No extra CCS is required for Global Ambition scenario

The two figures below display these changes:





As a consequense of the changes the two figures (figure 46 and 47) have been updated with the new numbers.

Figure 46

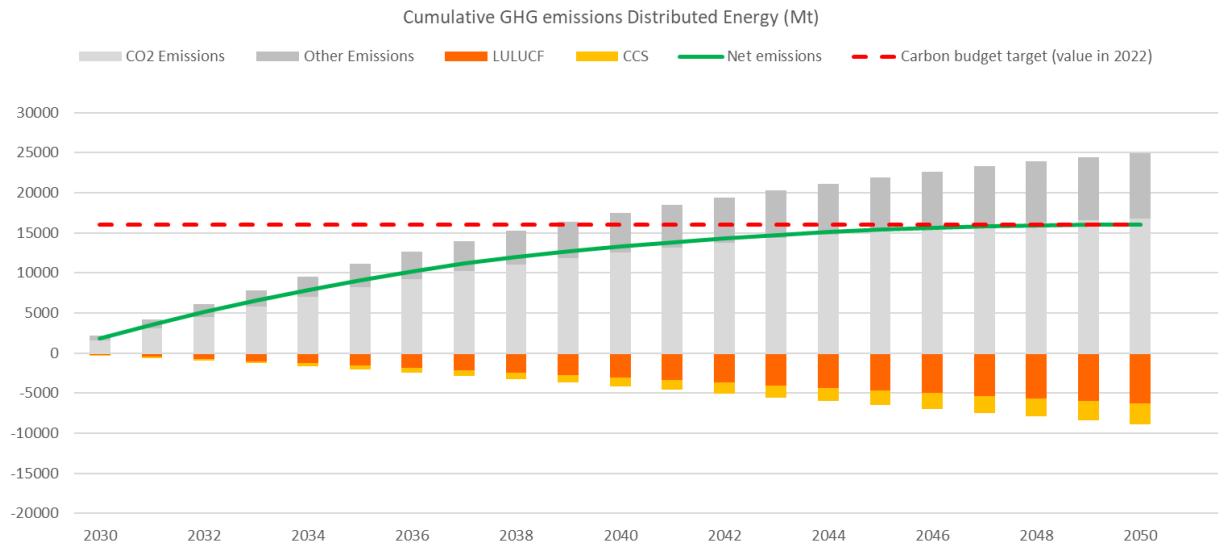
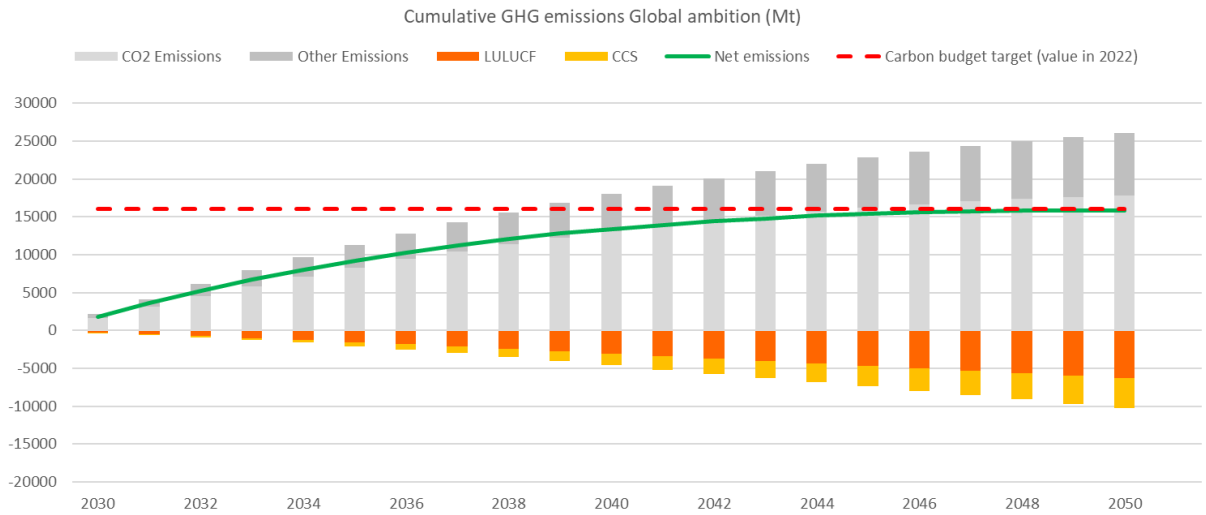



Figure 47